



Clinical challenges

Not everything is ischemic optic neuropathy

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ABSTRACT

A 71-year-old woman developed sudden, painful, decreased vision in the left eye accompanied by progressive instability. Initial examination revealed left optic disc edema, and macular optical coherence tomography confirmed the presence of intraretinal and subretinal fluid, as well as hyperreflective material under the retinal pigment epithelium. Subsequent investigations, including brain magnetic resonance imaging and a comprehensive serological analysis, ruled out infectious and autoimmune causes, further complicating the diagnostic picture. The patient's vision in both eyes continued to deteriorate, prompting empirical corticosteroid treatment. While the vision improved, the case took an unexpected turn with worsening neurological symptoms. Ultimately a brain biopsy was consistent with diffuse large B-cell lymphoma.

1. Case report

A 71-year-old woman with a history of systemic hypertension and dyslipidemia and no relevant ophthalmic history presented with sudden painful decreased vision in the left eye upon awakening. She also reported progressive gait instability during the previous week. She denied headaches or jaw claudication. Visual acuity was 20/25 in the right eye and 20/100 in the left eye. There was a left relative afferent pupillary defect. Left optic disc edema was observed (Fig. 1a), with no other findings of interest at the time. Optical coherence tomography (OCT) of the optic disc confirmed the left optic disc edema (Fig. 1b). Macular OCT revealed intraretinal and subretinal fluid in the left eye, along with hyperreflective material under the retinal pigmented epithelium (RPE). (Fig. 1c).

1. What do you think of the OCT peripapillary retinal nerve fiber layer (pRNFL) thickness and the macular findings?
2. Based on the patient's history and findings, what additional testing should be performed?

2. Comments by Dr. M Tariq Bhatti, MD

Even though nonarteritic anterior ischemic optic neuropathy (NAION) is the most common cause of unilateral optic disc edema in a patient over 50 years of age, as suggested by the title of this case; that is not always the situation. In this patient, however, NAION is certainly a high possibility given her age and risk factors of hypertension and dyslipidemia. We are not told the fellow eye cup-to-disc ratio, but if it is ≤ 0.2 that would also be consistent with NAION. Automated perimetry should be performed in all patients being evaluated for optic neuropathy, and if in this case there is an altitudinal defect that would support the diagnosis of NAION.²² Pain or periocular discomfort can occur in about 10 % of patients with NAION, but not pain with eye movements.⁷ Visual loss upon awakening in NAION is thought to be the result of nocturnal arterial hypotension, but recent studies have questioned this phenomenon.¹⁹

The increased pRNFL thickness is consistent with optic disc edema, but in itself does not discern the underlying etiology. A 5-line raster scan centered on the optic disc can provide a lot of additional information. For example, it can show vitreous traction indicative of vitreopapillary traction optic neuropathy or cellular infiltration within the neural tissue.²⁸

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The presence of intraretinal and/or subretinal fluid is not a disqualifying feature of NAION. In a recent study of 32 patients with NAION, nearly 50 % had retinal edema.⁶ Although not the case here, neuroretinitis is often associated with submacular fluid in the presence of macular exudates in a star pattern.¹³

A perplexing finding is the sub-RPE hyperreflective changes noted on

the OCT, which are not consistent with NAION. I am assuming this is new, but certainly one should consider the possibility that the finding was present prior to these most recent events and are unrelated. The most common reason for sub-RPE deposits is drusen but other entities to consider include retinal degeneration, infectious, inflammatory and neoplastic conditions such as retinitis pigmentosa, multifocal

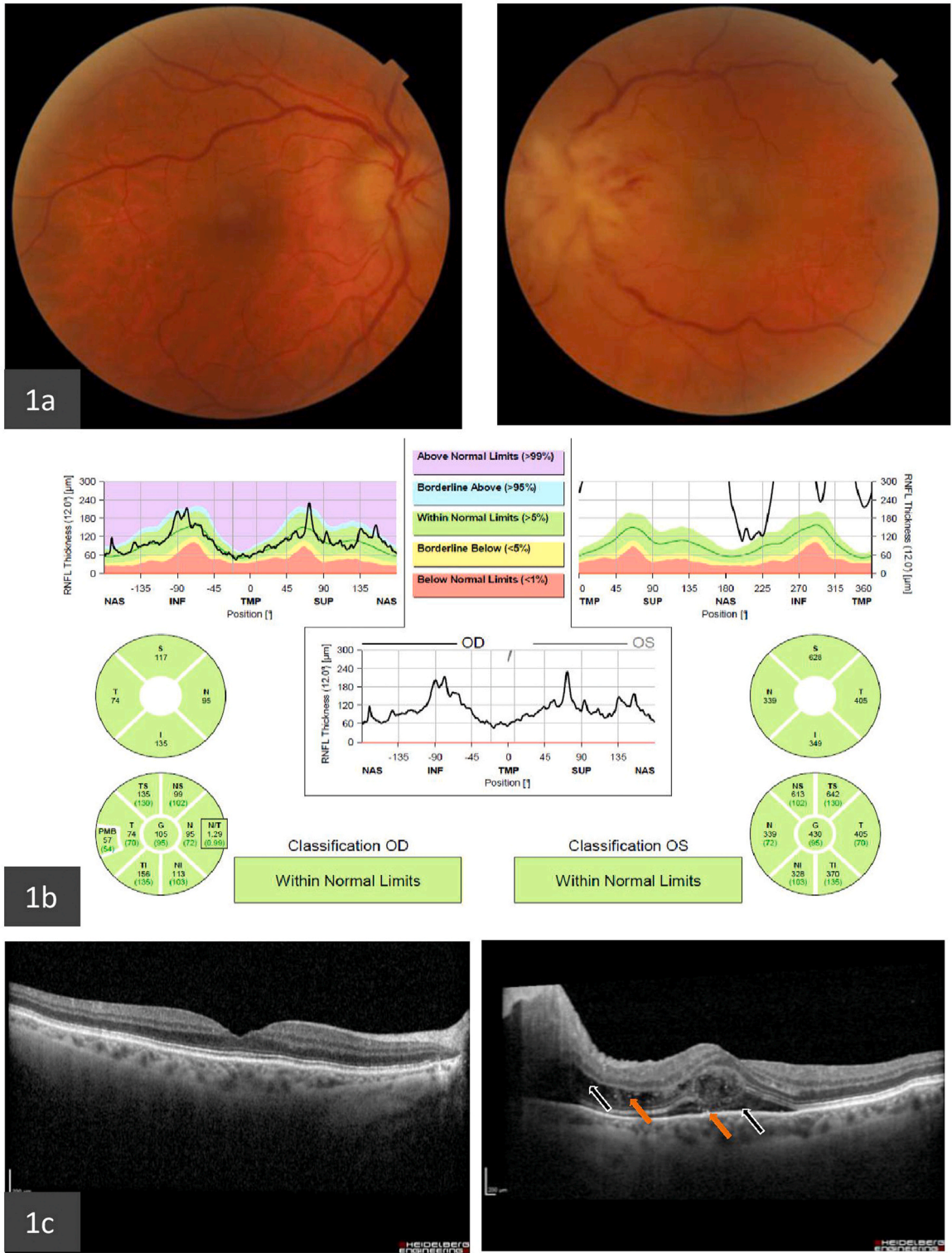


Fig. 1. a: Fundus photograph. Left optic disc edema. b: peripapillary retinal nerve fiber analysis. Increased thickness in the left eye c: Macular optical coherence tomography. Intraretinal and subretinal fluid (black arrows), as well as subretinal and sub-retinal pigment epithelial deposits of hyperreflective material (orange arrows).

choroiditis, birdshot retinochoroidopathy, syphilis, tuberculosis, sarcoidosis, and lymphoma.²⁹

Although she does not endorse headache or jaw claudication (and I am assuming she has no other symptoms such as scalp tenderness, anorexia, proximal joint pain, etc.), I would still screen her for giant cell arteritis with a complete blood count (CBC), erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP). I typically don't obtain a magnetic resonance imaging (MRI) in a patient whom I suspect has NAION.

I am not sure if the gait instability is an orthopedic, vestibular, or a neurological condition. Consultation with a neurologist should sort that out.

3. Case report continued

CBC, comprehensive metabolic panel, ESR and CRP were normal. Magnetic resonance imaging (MRI) of the brain and orbits with contrast demonstrated signal alteration in the globus pallidus (Figure 2). One week later the patient reported an onset of pain with eye movements in the right eye. Visual acuity was 20/50 in the right eye and 10/200 in the left eye. Keratic precipitates (KPs) were observed in both eyes, although there was no anterior chamber reaction. Funduscopy and OCT of the optic disc revealed bilateral optic disc edema. A serological analysis encompassing interferon-gamma release assay, human immunodeficiency virus antibodies, human T-lymphotropic virus I/II antibodies, Epstein-Barr virus antibodies, *Treponema pallidum* antibodies, *Borrelia burgdorferi* antibodies, onconeural antibodies (ie, Hu, Yo, Ri, Tr, CRMP5, amphiphysin, Ma2/Ma1, SOX1, ZIC4) and intracellular neuronal/glial antibodies (ie, GAD, GFAP, AK5) were all negative. Computed tomography (CT) of the chest, abdomen and pelvis was normal with absence of any discernible malignant lesions.

What would you do next?

4. Comments by Dr. Bhatti continued

There have been many new developments since the patient was initially seen, which underscores the importance of a follow-up examination and requires a reconsideration of the initial presumptive diagnosis of NAION.

Admittedly I would not have initially ordered the MRI, but now that it has been done we need to consider the causes of bilateral T2 hyperintense lesions involving the globus pallidus. Due to its high mitochondria content and high metabolic demand, the globus pallidus is vulnerable to infectious, metabolic-toxic, neurodegenerative, and vascular diseases. Specific etiologies of bilateral globus pallidus lesions including Wilson disease, carbon monoxide poisoning, Leigh disease, coronavirus disease 2019 (COVID-19), toxoplasmosis, pantothenate kinase-associated neurodegeneration (Hallervorden-Spatz syndrome), neurofibromatosis type 1, neuro-Béchet disease, methylmalonic acidemia, cocaine use, nonketotic hyperglycemia, ischemia, hypoxia, and hypercoagulation.²⁰ Rarely can bilateral globus pallidum lesion be associated with an intrinsic brain tumor such as a glioma or lymphoma.⁸

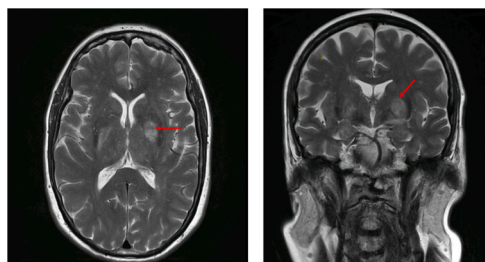


Fig. 2. Axial and coronal, T2-weighted magnetic resonance images. Hyperintense signal changes are present in the globus pallidus bilaterally (red arrow).

Focal or well-demarcated lesions involving a part of the basal ganglia, such as the globus pallidus, is highly suggestive of an infection or neoplasm.²⁰

The patient is experiencing pain with eye movements associated with new onset right optic disc edema, which is suggestive of optic neuritis.² Although it can happen, the lack of optic nerve enhancement on MRI is not typical for optic neuritis.⁵

Finally, the presence of anterior segment inflammation is not consistent with NAION and suggests an infectious, inflammatory, or neoplastic process. It would have been helpful to determine if the KPs were “mutton-fat” in appearance suggesting a chronic granulomatous inflammatory disease such as Vogt-Koyangi-Harada syndrome, sarcoidosis, tuberculosis, syphilis, or lymphoma.

Based on the list of possible diagnoses for each of the various clinical manifestations and by using a Venn diagram approach, it seems lymphoma would be the most likely unifying diagnosis. Therefore, it would be prudent to proceed with a lumbar puncture (LP) with opening pressure (OP) and cerebrospinal fluid (CSF) analysis.

5. Case report continued

A LP yielded an OP of 14 cmH₂O. Analysis of the CSF revealed a normal cell count of 5 nuclear cells per cubic millimeter (0–5), 0 red blood cells (0–1), a glucose level of 69 milligrams per deciliter (50–80), a protein level of 38.9 milligrams per deciliter (15–60), and a lactic acid level of 1.52 millimoles per liter (1–3). Additionally, CSF cytology was negative for malignant cells.

What is your differential diagnosis of visual loss?

6. Comments by Dr. Bhatti continued

Before discussing the possibilities of the visual loss, I would like to make a few comments regarding the process of analyzing CSF for a central nervous system (CNS) malignancy. Because the performance of a LP and interpretation of the CSF results is typically within the domain of the neurologist, I think it is fair to state that most ophthalmologists are not very familiar with the role of CSF analysis in suspected cases of neoplastic meningitis (NM). Here are a few key points to keep in mind:

- After a LP is performed, the CSF should be sent to the laboratory immediately. CSF that is left to sit for an extended period of the time (such as overnight) will result in cell lysis and result in a false negative result.

- A large volume (> 10 cc) of CSF should be collected for analysis.¹⁴

- The characteristic CSF profile of NM is a pleocytosis (60 % of cases), elevated protein (80 % of cases), decrease glucose (hypoglycorrhachia, 55 % of cases), and raised intracranial pressure (57 % of cases); however, these findings are non-specific and may be seen in inflammatory and infectious conditions.¹⁵

- Cytospin refers to centrifuge of the CSF to obtain a concentrated specimen that is then manually analyzed by light microscopy to characterize the cell composition and morphology (cytology). In comparison, flow cytometry refers to an automated technique that uses antibodies directed against cell-surface proteins to characterize the immunophenotype and morphology of each cell.

- The yield of positive cytology following a second lumbar puncture after the first negative lumbar puncture is increased by 25 %.¹⁵

- CSF cytology has a sensitivity of 2–32 % in CNS lymphoma.²⁵ The diagnostic value of flow cytometry is twofold greater than cytospin for CNS hematologic malignancies.⁴ The combination of both increases the diagnostic yield.¹²

- CSF molecular biology, molecular genetics and flow cytometry using a cell stabilizer medium have a greater sensitivity of detecting malignant cells.³

Given the eye pain with eye movements, I would repeat the MRI (brain and orbits with contrast) because I have personally seen several patients in which the initial MRI did not show optic nerve enhancement,

but the subsequent study did. In addition, I would check aquaporin 4 immunoglobulin IgG and myelin oligodendrocyte glycoprotein IgG.

7. Case report continued

One week later, the patient presented to the emergency room with a decline in visual acuity in the right eye that had been progressing over several days. Visual acuity measured 20/400 in the right eye and counting fingers at 6 in. in the left eye. There were KPs, optic disc edema with peripapillary hemorrhages, and vitritis in both eyes. A fluorescein angiography revealed no signs of retinal vasculitis. Considering the significant deterioration in her vision, empirical corticosteroid treatment (60 mg every 24 h) was initiated.

What is happening?

8. Comments by Dr. Bhatti continued

What is happening to this patient is a very good question, and I would like to know myself! On a basic level what she has is a rapidly progressive bilateral optic neuropathy.

I understand that sometimes an empiric trial of corticosteroids needs to be considered in some patients, but I personally hesitate to employ this strategy for several reasons. First, I don't like the thought of treating an unknown condition. Second, clinical improvement, which is a good thing, can occur in the setting of both an inflammatory and neoplastic process so it does not suggest a particular diagnosis. Third, and most importantly, I worry that corticosteroids can worsen an infectious process that I was not considering.

Although there is a low likelihood of this being GCA with negative inflammatory markers, I would still recommend a temporal artery biopsy.²³ I would also obtain an MRI of the brain and orbits with contrast.

9. Case report continued

Over the subsequent weeks there was significant improvement in visual acuity (20/32 right eye and 20/63 left eye); however, she manifested a notable functional decline, reporting symptoms of forgetfulness and gait instability. Another LP was performed with normal OP. The CSF was normal with 0 nuclear cells (0–5), 53 red blood cells per cubic millimeter (0–1), a glucose level of 60 milligrams per deciliter (50–80), and a protein level of 66.5 milligrams per deciliter (15–60). CSF gram stain and culture was negative for microorganisms. Furthermore, CSF cytology once again was negative for malignant cells.

What are the next steps?

10. Comments by Dr. Bhatti continued

It seems reassuring that the second LP is negative, but that does not completely exclude a CNS process. If there are still vitreous cells, I would consult a vitreoretinal surgeon for a vitreous biopsy. Also, as mentioned earlier, I would repeat the MRI (brain and orbits with contrast) and see her back sooner than later before deciding on what to do with the prednisone.

An emergent consultation with a neurologist is needed because it is very evident that she not only has a rapidly progressive optic neuropathy, but also an acute progressive neurological disorder.

11. Case report continued

Additional blood test indicated the presence of IgG for cytomegalovirus and toxoplasmosis, with no other noteworthy findings. Repeat onconeural antibody testing was negative. Two weeks later, the patient experienced a worsening of vision, with a visual acuity of 20/60 in the right eye and 10/200 in the left eye. There was no vitritis. It was decided to initiate a descending regimen of topical and systemic corticosteroids. Subsequently, a paracentesis of the anterior chamber was

performed for the molecular detection of the myeloid differentiation factor 88 (MYD88) mutation, followed by a vitreous biopsy. Both procedures were performed separately to mitigate the risk of ocular hypotension. The analysis of MYD88 by polymerase chain reaction in the aqueous humor sample was negative. The examination of the vitreous biopsy revealed the presence of atypical lymphocytes, but the quantity was insufficient for flow cytometry analysis. A whole-body positron emission tomography (PET)-CT scan was negative.

Would the findings in the vitreous alter your management?

12. Comments by Dr. Bhatti continued

Now, I am highly suspicious this is intraocular-CNS lymphoma that needs to be cytopathologically or histopathologically confirmed. I would speak directly with the pathologist who interpreted the vitreous biopsy to get his or her opinion on the significance of the “atypical” lymphocytes, as well as contact the vitreoretinal surgeon to discuss whether another vitreous biopsy could be done. In addition, as I have stated before, I would repeat the MRI.

13. Case report continued

Two months after the onset of symptoms, a new brain MRI revealed patchy and confluent infiltrative lesions distributed bilaterally and almost symmetrically, involving both supra- and infratentorial regions (Figure 3). The uptake pattern appeared homogeneous, with no signs of necrosis. A subsequent brain biopsy confirmed the diagnosis of primary brain lymphoma that immunohistochemistry showed to be a diffuse large B-cell lymphoma. Following an evaluation by the hematology department, the possibility of curative treatment was ruled out. The patient's clinical progression was characterized by significant and progressive functional and visual deterioration, ultimately leading to their death 3 months after the initial symptom presentation.

14. Discussion

CNS lymphoma represents an uncommon, but exceptionally aggressive, form of non-Hodgkin lymphoma (NHL), which can be broadly divided into 2 primary subtypes: primary CNS lymphoma (PCNSL) and secondary CNS involvement by systemic lymphoma. PCNSL, a rare variant of extranodal NHL, infiltrates the neuraxis, encompassing the orbit, brain, leptomeninges, and spinal cord. PCNSL represents 1 % of all NHLs and 3 % of all tumors, with an incidence rate of approximately 0.5 per 100,000 per year.¹¹ Histologically, 95 % of PCNSL are diffuse large B-cell lymphoma, the remaining 5 % being low-grade B-cell lymphoma, T-cell lymphoma, and Burkitt lymphoma.¹⁶

Up to 80 % of patients diagnosed with PCNSL may exhibit focal neurologic deficits. Cognitive impairment is the most prevalent initial presentation, followed by gait disturbances, focal neurological deficits, signs of elevated intracranial pressure, and seizures.¹⁶

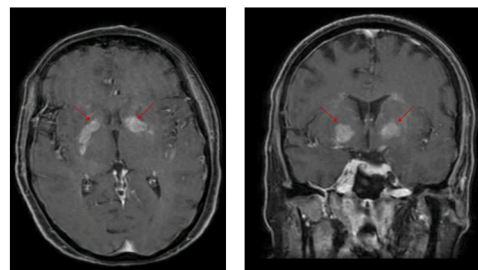


Fig. 3. Axial and coronal, T1 weighted, post-contrast magnetic resonance images. There is patchy and confluent infiltrative lesions distributed bilaterally and almost symmetrically, involving both supra- and infratentorial regions (red arrows).

Vitreoretinal lymphoma (VRL) is an uncommon manifestation of PCNSL, often involving the retina, vitreous, or optic nerve head. It can manifest in isolation or more frequently in conjunction with parenchymal brain lymphoma. At the time of diagnosis, approximately 15 % to 25 % of patients with PCNSL may present with ocular dissemination;¹⁷ however, a careful distinction between systemic lymphoma with orbital involvement is essential.

Diagnosing VRL can be challenging due to its similarity to other forms of uveitis. Anterior segment findings associated with VRL include KPs, pseudo-hypopyon, iris and angle infiltration.²¹ The characteristics of KPs have been studied with *in vivo* confocal microscopy aiming to develop biomarkers for PCNSL.^{18,21} Nevertheless, the limited sample size of cases hinders the ability to draw any conclusive results.

In most patients fundus examination typically reveals vitreous inflammation, varying in intensity from mild to severe. Contrary to the extent of inflammation observed during clinical examination, visual acuity is typically better than expected in the majority of cases. Lymphoma cells in the vitreous are uniform, larger than inflammatory cells and don't cluster with reactive cells, creating a visual effect like the "aurora borealis." VRL patients often exhibit creamy lesions with yellowish infiltrates deep in the retina or RPE, forming a distinctive "leopard skin" pigmentation. Other described retinal findings in VRL includes isolated subretinal lesions, exudative retinal detachment, RPE atrophy with subretinal fibrosis, and disciform scarring at the macula.²⁷

Optic nerve infiltration (ONI) as the initial manifestation of PCNSL is rare. Ahle and coworkers conducted a review of 752 patients with PCNSL, revealing that only 7 patients exhibited lymphomatous ONI.¹ Among these 7 patients, only 2 presented with ONI at the initial time of diagnosis. Lymphomatous ONI manifested subacutely, featuring severe visual loss, a normal fundus appearance, and optic nerve contrast enhancement, with or without optic nerve thickening observed on MRI.

In our case, we believe that macular OCT observations, particularly the subretinal hyperreflective lesions, were related to lymphomatous infiltration, which was probably responsible for the visual deterioration. In this regard, Pichi and coworkers conducted a retrospective review of 181 eyes from 115 patients with biopsy-proven VRL who underwent OCT imaging at presentation, and found that lymphomatous infiltration, manifested as hyperreflective material, was identified in three main regions: the sub-RPE space, the subretinal space, and the intraretinal space.²⁴ Furthermore, in this study, isolated sub-RPE involvement was the most prevalent, followed by a combination of sub-RPE and subretinal lesions. Sub-RPE macular lesions were associated with a worse visual acuity, when compared with those cases with no hyperreflective material in the macular or only sub-RPE lesions.

A definitive diagnosis of PCNSL with ocular involvement is usually established through histopathological examination of brain biopsy, vitrectomy, or CSF cytology. Additionally, flow cytometry studies can be employed to determine the type and clonality of the lymphocytes.¹⁰ The MYD88 protein transduces signals to nuclear factor kappa light chain enhancer of activated B-cells factor, influencing lymphoma development. MYD88 mutation can be detected in samples of aqueous humor, vitreous humor and CSF. In a sample of 18 eyes with biopsy-confirmed or clinically diagnosed VRL, Demirci and coworkers found that 83 % of patients had detection of aqueous humor-derived MYD88 L265P mutation.⁹ Molecular analysis of this mutation may represent a less invasive diagnostic tool in these patients, although its usefulness in everyday clinical practice remains to be seen.

Given the non-specific nature of symptoms and the typical initial positive response to corticosteroid treatment, it comes as no surprise that there is often a considerable delay in diagnosing VRL, extending up to 40 months in certain cases.²⁶ Therefore, maintaining a high clinical suspicion is crucial, and emerging diagnostic tools such as MYD88 mutation analysis in aqueous humor may emerge as invaluable assets for early detection in these patients. Enhancing our diagnostic arsenal with these tools is pivotal for timely intervention and improved patient outcomes.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Ahle G, Touitou V, Cassoux N, et al. Optic Nerve Infiltration in Primary Central Nervous System Lymphoma. *JAMA Neurol.* 2017;74(11):1368.
- Beck, Cleary RW, Anderson PA, et al. A randomized, controlled trial of corticosteroids in the treatment of acute optic neuritis. The Optic Neuritis Study Group. *N Engl J Med.* 1992;326(9):581–588.
- Bento LC, Correia RP, Alexandre AM, et al. Detection of Central Nervous System Infiltration by Myeloid and Lymphoid Hematologic Neoplasms Using Flow Cytometry Analysis: Diagnostic Accuracy Study. *Front Med (Lausanne).* 2018;5:70.
- Bromberg JE, Breems DA, Kraan J, et al. CSF flow cytometry greatly improves diagnostic accuracy in CNS hematologic malignancies. *Neurology.* 2007;68(20):1674–1679.
- Bursztyn LLC, De Lott LB, Petrou M, Cornblath WT. Sensitivity of orbital magnetic resonance imaging in acute demyelinating optic neuritis. *Can J Ophthalmol.* 2019;54(2):242–246.
- Chapelle AC, Rakic JM, Plant GT. The Occurrence of Intraretinal and Subretinal Fluid in Anterior Ischemic Optic Neuropathy: Pathogenesis, Prognosis, and Treatment. *Ophthalmology.* 2023;130(11):1191–1200. <https://doi.org/10.1016/j.ophtha.2023.07.015>. Epub 2023 Jul 20. PMID: 37479117.
- Characteristics of patients with nonarteritic anterior ischemic optic neuropathy eligible for the Ischemic Optic Neuropathy Decompression Trial. *Arch Ophthalmol.* 1996;114(11):1366–1374. <https://doi.org/10.1001/archophth.1996.01100140566007>. PMID: 8906027.
- Da Silva AJ, Castro Pinheiro Gomes FE, Barros Pimentel Tenório GM, Pinto Almeida LM. Diffuse Large B-Cell Lymphoma in Bilateral Basal Ganglia: A Rare Case Report. *Cureus.* 2022;14(12), e32743.
- Demirci H, Rao RC, Elnor VM, et al. Aqueous Humor-Derived MYD88 L265P Mutation Analysis in Vitreoretinal Lymphoma: A Potential Less Invasive Method for Diagnosis and Treatment Response Assessment. *Ophthalmol Retin.* 2023;7(2):189–195.
- Ferreri AJM, Calimeri T, Cwynarski K, et al. Primary central nervous system lymphoma. *Nat Rev Dis Prim [Internet].* 2023;9(1):29. Jun 15 (<https://www.nature.com/articles/s41572-023-00439-0>).
- Fox CP, Phillips EH, Smith J, et al. Guidelines for the diagnosis and management of primary central nervous system diffuse large B-cell lymphoma. *Br J Haematol [Internet].* 2019;184(3):348–363. Feb 1 [cited 2023 Nov 14] (<https://onlinelibrary.wiley.com/doi/full/10.1111/bjh.15661>).
- French CA, Dorfman DA, Shaheen G, Cibas ES. Diagnosing lymphoproliferative disorders involving the cerebrospinal fluid: increased sensitivity using flow cytometric analysis. *Diagn Cytopathol.* 2000;23:369–374.
- Ganatra S, Panchal B, Pathengay A, Sachdeva V. Differentiating Occult Neuroretinitis and Non-Arteritic Anterior Ischaemic Optic Neuropathy: Clinical and Optical Coherence Tomography Characteristics. *Neuroophthalmology.* 2023;47(4):208–217.
- Glantz MJ, Cole BF, Glantz LK, et al. Cerebrospinal fluid cytology in patients with cancer: minimizing false-negative results. *Cancer.* 1998;82:733–739.
- Gleissner B, Chamberlain MC. Neoplastic meningitis. *Lancet Neurol.* 2006;5(5):443–452.
- Green K, Munakomi S, Hogg J.P. Central Nervous System Lymphoma. *Curr Diagn Pathol [Internet].* 2023 Aug 13 [cited 2023 Nov 14];3(1):45–52. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK545145/>.
- Grimm SA, McCannel CA, Omuro AMP, et al. *Prim CNS Lymphoma intraocular Involvement PCNSL Collab Group Rep.* 2008.
- Guan W, Zhang Y, Zhang X, Peng X. In Vivo Confocal Microscopy of Keratic Precipitates in Vitreoretinal Lymphoma. *Ocul Immunol Inflamm.* 2022;1–6.
- (a) Hayreh SS, Podhajsky P, Zimmerman MB. Role of nocturnal arterial hypotension in optic nerve head ischemic disorders. *Ophthalmologica.* 1999;213(2):76–96.(b) Labowsky MT, Rizzo 3rd JF. The Controversy of Chronotherapy: Emerging Evidence regarding Bedtime Dosing of Antihypertensive Medications in Non-Arteritic Anterior Ischemic Optic Neuropathy. *Semin Ophthalmol.* 2023;38(1):99–104.
- (a) Hegde AN, Mohan S, Lath N, Lim CC. Differential diagnosis for bilateral abnormalities of the basal ganglia and thalamus. *Radiographics.* 2011;31(1):5–30. (b) Parekh MA. Bilateral globus pallidus lesions. *BMJ Case Rep.* 2021;14(2), e241088. <https://doi.org/10.1136/bcr-2020-241088>. PMID: 33547109; PMCID: PMC7871252.(c) Kulick-Soper CV, McKee JL, Wolf RL, et al. Pearls & Oy-sters: Bilateral globus pallidus lesions in a patient with COVID-19. *Neurology.* 2020;95(10):454–457.

21. Mahendradas P, Srinivasan T, Kawali A, et al. Floral Pattern of Keratic Precipitates in Vitreoretinal Lymphoma on In Vivo Confocal Microscopy. *Ocul Immunol Inflamm.* 2021;29(3):500–506.
22. Miller NR, Arnold AC. Current concepts in the diagnosis, pathogenesis and management of nonarteritic anterior ischaemic optic neuropathy. *Eye (Lond).* 2015; 29(1):65–79.
23. Parikh M, Miller NR, Lee AG, et al. Prevalence of a normal C-reactive protein with an elevated erythrocyte sedimentation rate in biopsy-proven giant cell arteritis. *Ophthalmology.* 2006;113(10):1842–1845.
24. Pichi F, Dolz-Marco R, Francis JH, et al. Advanced OCT Analysis of Biopsy-proven Vitreoretinal Lymphoma. *Am J Ophthalmol.* 2022;238:16–26.
25. Scott BJ, Douglas VC, Tihan T, Rubenstein JL, Josephson SA. A systematic approach to the diagnosis of suspected central nervous system lymphoma. *JAMA Neurol.* 2013; 70(3):311–319.
26. Soussain C, Malaise D, Cassoux N. Primary vitreoretinal lymphoma: a diagnostic and management challenge. *Blood.* 2021;138(17):1519–1534.
27. Venkatesh R, Bavaharan B, Mahendradas P, Yadav NK. Primary vitreoretinal lymphoma: prevalence, impact, and management challenges. *Clin Ophthalmol.* 2019; 13:353–364.
28. (a) Weng CY, Khimani KS, Foroozan R, Gospe 3rd SM, Bhatti MTA. sticky situation. *Surv Ophthalmol.* 2018;63(5):736–744 (Sep-Oct).(b) Fortes BH, Dalvin LA, Hogan WJ, et al. Isolated T-cell acute lymphoblastic leukemic optic disc infiltration. *Am J Hematol.* 2021;96(12):1717–1718. Dec 1.
29. (a) Zaharova E, Sherman J. The use of SD-OCT in the differential diagnosis of dots, spots and other white retinal lesions. *Eye Brain.* 2011;3:69–80. <https://doi.org/10.2147/EB.S23208>. PMID: 28539776; PMCID: PMC5436184.(b) Kuntz CA, Jacobson SG, Cideciyan AV, et al. Sub-retinal pigment epithelial deposits in a dominant late-onset retinal degeneration. *Invest Ophthalmol Vis Sci.* 1996;(9). Aug; 37.